

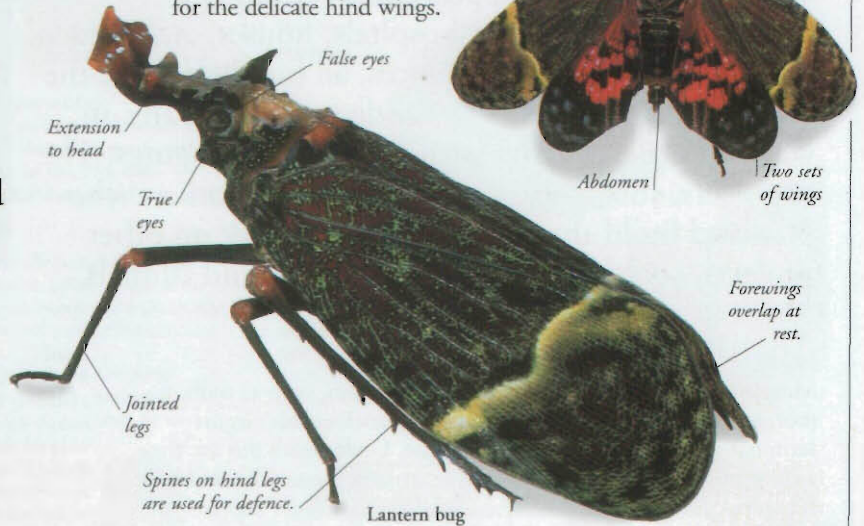
BUGS



THE WORD BUG is often used to describe any crawling insect or a disease-causing germ. The true bugs are a group of insects that have long feeding tubes specially adapted for sucking fluids out of plants and animals. Bugs, such as shield bugs, are often brightly coloured, and, as a group, they are remarkably varied in shape. There are about 55,000 species of bug, including large solitary insects, such as giant water bugs and cicadas, and tiny creatures, such as scale insects, bedbugs, and aphids. It is the smaller bugs, such as greenfly and leaf hoppers, that create problems for farmers because of the severe damage they cause to crops.

Features of a bug

All bugs have specialized mouthparts with cutting implements for piercing, and needle-like sucking tubes held within a protective sheath. Some bugs, such as lantern bugs, have their membranous wings exposed when at rest; others have forewings that are partially thickened and used not for flight, but as a protective cover for the delicate hind wings.



Parthenogenesis

Aphids such as greenfly and blackfly, multiply rapidly, because they can reproduce without mating. Females produce a succession of identical female offspring from unfertilized eggs, each of which later produces more of the same. This is called parthenogenesis.

Reproduction

Bugs attract a mate in many ways, such as giving off scent, or vibrating the surface of water. Male cicadas attract females with their loud song, produced by drum-like organs on the abdomen. During mating, male and female bugs are often attached for hours. Females usually lay hundreds of eggs. These hatch into nymphs – tiny versions of their parents – and moult many times before reaching adult size.

Shield bugs

Shield bugs are found virtually worldwide. They are also called stink bugs, as they can give off a bad smell. Females protect their eggs and young from attack.



Young shield bug nymphs being guarded by their mother.

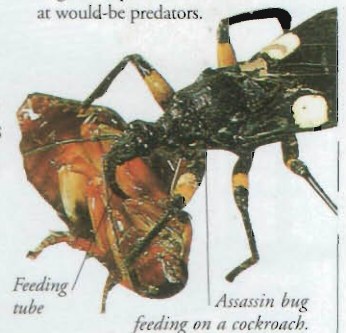
Feeding

Bugs use their mouthparts to cut a hole in their food and pierce the soft parts inside. They inject enzymes and digestive juices through a pair of tiny tubes to break down solids and suck up the resulting fluids. In this way, predatory bugs, such as assassin bugs, can suck their victims dry. Bedbugs are parasites that suck the blood of birds and mammals, including humans. Some bugs feed only on plant juices.



Assassin bugs

Assassin bugs are carnivores. Most prey on other invertebrates, such as millipedes. Some steal prey already caught in spiders' webs. Assassin bugs can squirt toxic saliva at would-be predators.



Feeding tube

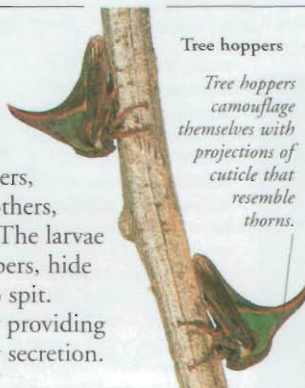
Assassin bug feeding on a cockroach.

Leaf hoppers

Leaf hoppers are herbivores. They are often considered pests as they cut holes in the leaves of plants, such as cotton plants, to suck out the sap, thereby weakening the plants.

Defence

Small bugs face many enemies from ladybirds to birds. To deter would-be attackers, bugs have evolved a range of defences. Some bugs, such as tree hoppers, have developed elaborate camouflage; others, such as stink bugs, give off bad smells. The larvae of spittle bugs, also known as frog hoppers, hide within a frothy substance called cuckoo spit. Aphids employ ants to protect them by providing their guardians with a nutritious sugary secretion.



Tree hoppers

Tree hoppers camouflage themselves with projections of cuticle that resemble thorns.

Water bugs

Some bugs live in water. Pond skaters skim over water on their dainty legs, while water boatmen dart below the water using paddle-shaped limbs. Underwater bugs come to the surface to breathe, or carry around an air bubble.



Pond skater

RED-BANDED LEAF HOPPER

SCIENTIFIC NAME

Graphocephala coccinea

ORDER

Homoptera

FAMILY

Cicadellidae

DISTRIBUTION

Eastern USA and eastern Canada

HABITAT

Meadows and gardens

DIET

Plant juices

SIZE

Length 8–11 mm (0.4–0.5 in); wingspan 12–16 mm (0.5–0.6 in)

LIFESPAN

Adults: up to 4 months

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PARASITES

PLANTS, DEFENCE